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Sandakan and Semporna Basin Exploration Potential Evaluation Via Scalable Dynamic Sub-Surface Evaluation Workflow

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The project was intended to unveil hidden potential of the East Sabah Basin in the Borneo in terms of the hydrocarbon opportunity. Considering the huge success in the other parts of Borneo such as Sarawak, NW Sabah basin, Brunei, Kalimantan etc., the East Sabah basin could be a potential area to further investigate. In this project, Halliburton has taken up a new approach called "Dynamic Subsurface Evaluation", integrating some state of art digital technologies to expedite the whole evaluation process from seismic interpretation to prospect generation and risking. The notable technology used in this project is "Scalable Earth Modeling" which is a grid-less modeling approach reducing the challenges of the resolution of the model augmented by the Neflex® solution to leverage the regional understanding from the proprietary geology database of Neflex®. Additionally, some detailed workflows like full scale 3D Petroleum Systems Modeling have also been adopted to de-risk the play and prospect to uncover the potential of the exploration opportunity.

Analysis of techno-stratigraphy and seismic data in the Sandakan Basin, offshore eastern Sabah, Malaysia, spanning several decades of exploration since the 1970s, with some minor hydrocarbon discoveries. Out of the twenty-five (25) wells drilled in the basin until 2015, only seven (7) have yielded hydrocarbon discoveries, all occurring between 1970 and 2000. Through the drilling of these wells there are mainly three main proven play types have been identified:

- i. The "Tungku play," occurring from Early to Middle Miocene, targets reservoirs within the Tanjong Formation equivalents in the Segama Group, deposited as part of the synrift sequence.
- ii. The "Sebahat play," spanning from Middle to Late Miocene, features reservoirs within the Sebahat Formation, deposited as the part of post-rift sequence.
- iii. The Ganduman Play, Late Miocene play which is characterized by prograding deltaic shoreface and shelf sequences, advancing southeastwards from an uplifting hinterland in central and northern Sabah.

The Sandakan Basin has experienced several extensional and compressional deformation events, resulting in major erosional unconformities, notably during the Middle and Late Miocene. These unconformities, caused by compression and faulting, are responsible for trap formation but also pose risks to trap integrity and preservation.

Keywords: Scalable Earth Modeling, Neflex®, 3D Petroleum Systems Modeling, Play Chance Mapping

Introduction

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Objectives

The objective of the study is aimed at evaluating the subsurface insights in Sandakan and Semporna basins through a holistic approach of play to prospect assessment. A dynamic subsurface evaluation has been carried out to generate play chance maps and translating to prospect generation, volumetric, and assessment in order to help evaluate the basin potential. The key challenges were to handle 2D seismic data coverage in 17,000 sq.km. with 37 exploration wells and uncertainty pertaining to interpretation and velocity modeling. The study area covers around 17,000 sq. km. and contains approximately 19 exploration wells and 18 side tracks.

Methodology

The end-to-end exploration workflow solution ranges from seismic interpretation to geological model building followed by detailed gross depositional map generation and 3D facies model building using an innovative approach of grouped proportion curve model. The project methodology also includes full-scale 3D Petroleum Systems Modeling and Play Fairway Mapping to screen the play area based on the standard Exxon Risk model. This exploration risk model involves both the geological indicator and the confidence factor. A detail GDE map creation approach has been taken into account, integrating data from seismic reflection, thickness maps, well data etc. A region-based Vertical Proportion Curve (VPC) is created using

Tau Model to convert the VPC into a 3D Gross Depositional Environment Model with control from the background GDE maps.

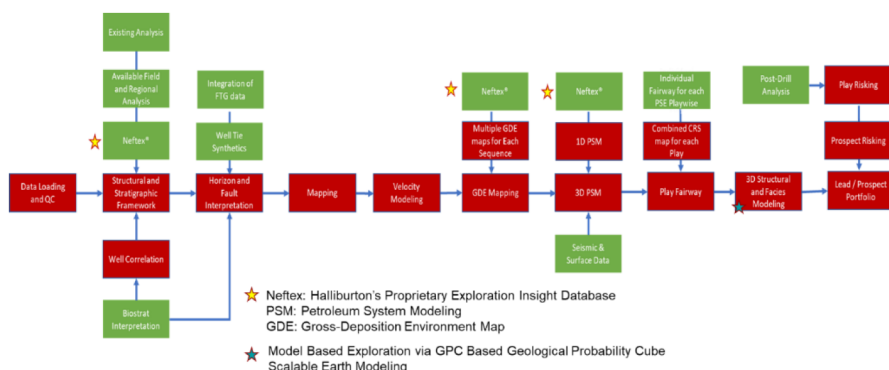


Figure 1—The Adopted Workflow for the Dynamic Subsurface Evaluation

Basin scale interpretation has been carried out to capture individual play concept in integration with the Neflex® proprietary stratigraphic framework. The objective of the project is to incorporate all the proven plays along with deeper speculative plays. The detailed evaluation also managed to unveil a potential Miocene Carbonate play which is quite well established in the other part of Borneo though never explored in the Eastern part of Sabah.

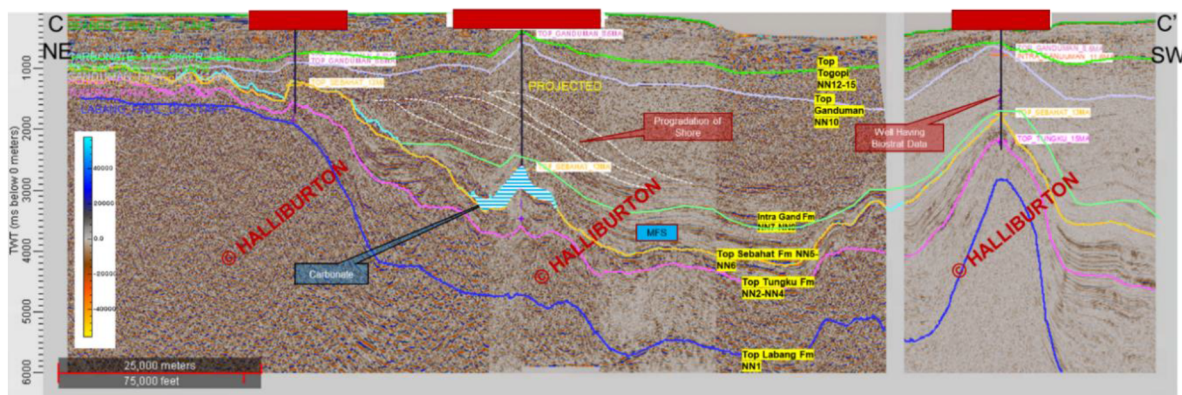


Figure 2—The Regional Interpretation for Play Concept Evaluation

The project work also highlighted one of the common challenges related to the data integration beyond the study area. The study area doesn't have a depocenter included to have a mature source rock which has been addressed by integration of regional data from NefTex® into the Petroleum Systems Model by testing the scenario of hydrocarbon charging from the outside source pod into the East Sabah Basin area.

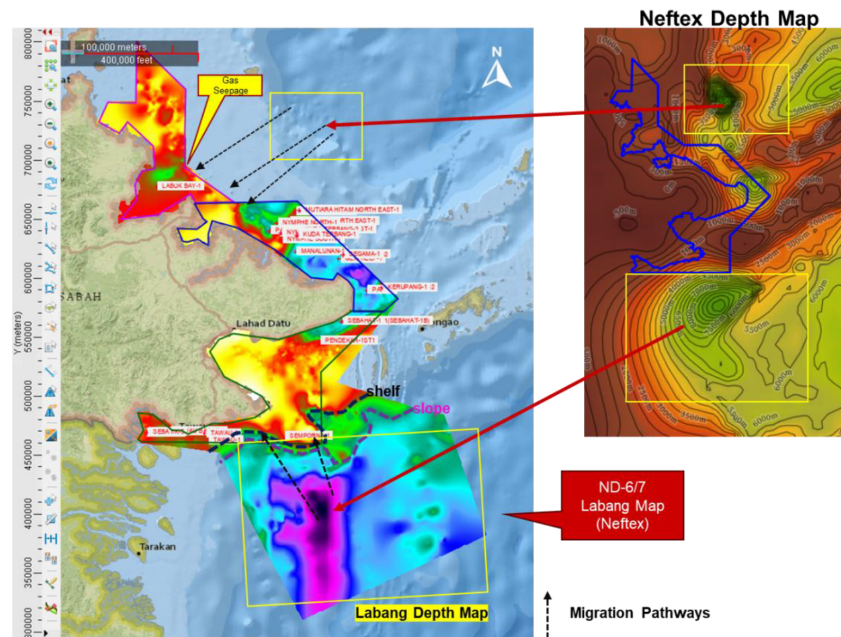


Figure 3—The Value derived from the Regional Context Integration to Evaluate the Petroleum Systems Elements for Play Risking Study

Results & Observations

The below key components were evaluated at the play scale:

- Major Play: Early-Late Miocene Shallow Marine to Turbidite Play, Early-Mid Miocene Carbonate Play, Deeper Oligocene play
- Structural Elements: Multiple Extension and Late Structural Inversion, Complex Fault System (inverted structures at deeper and shallow levels, growth faults)
- Complexity: Depositional system is influenced by complex tectonic events through time from extension to inversion and structural reactivation

GDE Map creation constraints:

The GDE mapping workflow begins at the mega-regional scale to capture the major paleogeographic trends and active tectonic features. The maps are then refined at higher resolution to incorporate the project specific constraint from seismic data and wells. In this study the following data constraints were used to construct all GDE maps:

1. Neflex® Predictions regional GDEs – these maps cover the entirety of South East Asia and capture the mega-regional depositional trends.
2. Paleogeographic reconstructions – the regional concept based GDE maps from Dr. Allagu Babaguru were used to map the position of key paleogeographic features and facies trends.
3. Pre-existing studies (e.g. ISIS study (2005), Petronas Milestone Review 2 (2023)) – relevant maps from these studies were tied to the stratigraphic framework and used to support and/or refine facies trends and the morphology of certain depositional features (e.g. deep-water fans).
4. Well constraint – the well dataset available for this study was reviewed to capture the proven petroleum systems elements and depositional facies for each lithostratigraphic unit. The facies information from the well was then tied to each specific GDE surface and used to constrain and support the GDE map.
5. Seismic constraint – morphological features (e.g. shelf break, clinoforms, carbonate build-ups etc) were interpreted from the 2D seismic lines and then translated as polylines to constrain the GDE maps.

- Sequence isopachs and structural depth maps – these are derived from the seismic interpretation and used to review trends in the evolution of accommodations space between the different GDE map surfaces.

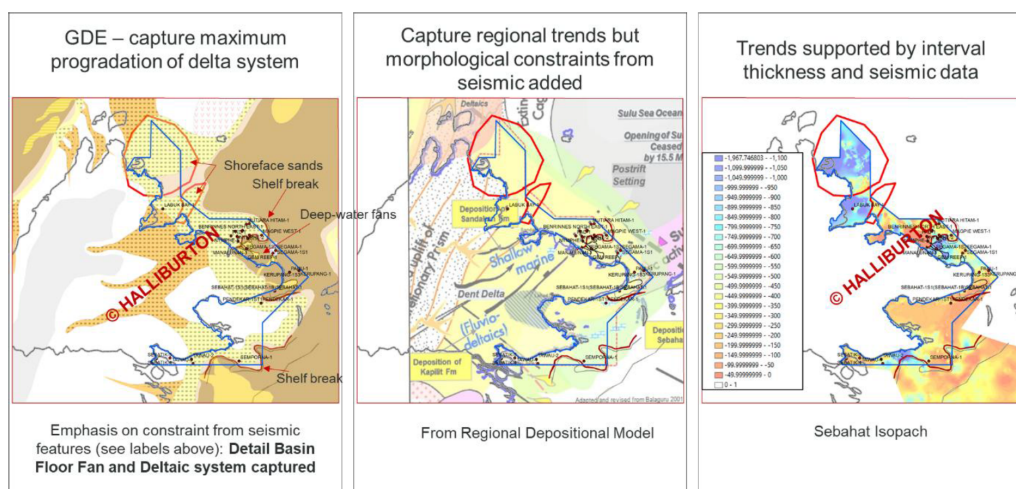


Figure 4—The Standard Integrated Approach adopted for GDE Map creation utilizing Seismic to Geological Data

Scalable Earth Modeling: Once the GDE Maps are prepared, an innovative Grid-less Modeling approach has been adopted to bring the scalability in the regional model. The biggest challenge in such regional models is grid-resolution which forces the modeler to compromise between the extent of the model and the resolution detailing. In order to address the resolution challenge, the Scalable Earth model builds a grid-less model which can be used with the zoom in functionality to get a deep down in any part of the model with intended resolution without any requirement of building a new model. The Scalable Earth model has been further used in the Petroleum Systems Modeling workflow covering the entire sediment pile from Basement to top.

Petroleum Systems Modelling is an essential tool in the evaluation of conventional and unconventional exploration opportunities and in reservoir development projects. Petroleum Systems Modelling studies address the characterization of source rock and fluids; the generation, migration, accumulation and alteration of hydrocarbons in a sedimentary basin; and the geological, physical and chemical processes that govern them.

Petroleum systems analyses thus include all the major geologic disciplines involved in petroleum resource assessments: reservoir, seal, trap and charge. However, the only framework in which all of these elements are usually considered together and in the context of basin evolution is in a basin modeling study.

When optimally conceived and executed, a basin model is an integration and learning tool, involving specialists and generalists from a variety of different disciplines. These might include pore pressure prediction, structural geology, seismic interpretation, sedimentology, seal analysis, tectonic analysis, petrophysics, petroleum engineering, diagenesis, and geochemistry.

The detailed 3D geological model was built through a grid-less approach which included regional GDE information to prospect scale, well, and seismic insights without losing any resolution at any stage. This evergreen 3D earth model also serves towards 3D Petroleum Systems Modeling to assess the play risk and prospect scale de-risking. As an outcome of the project work, a full-fledged lead inventory was created with portfolio scale risking and volumetrics. Hence, the dynamic subsurface evaluation covers an integrated workflow from interpretation to geological modeling to Petroleum Systems Modeling & Play Chance Mapping, ranging to lead/prospect scale risking & volumetrics supported by a digital innovative approach of "Scalable Earth Modeling".

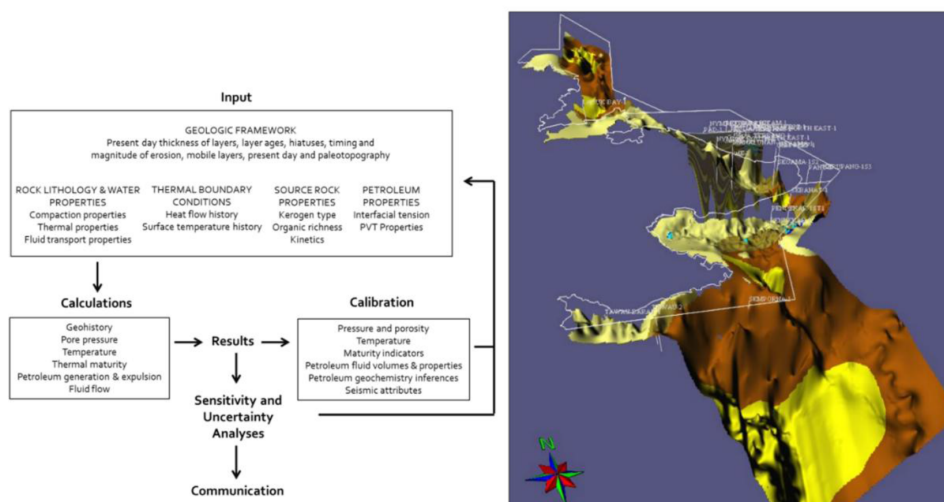


Figure 5—The Scalable Earth Model integration with Petroleum Systems Modeling for the full-scale 3D PSM

The 3D Geological Model has been used to run the full-scale 3D Petroleum Systems Model including the Thermal, Pressure, and Migration Models. The model has been calibrated using Pressure, Temperature, and VRo data. The final results from migration model have been further included in the Play Chance Mapping workflows. One of the benefits of such evergreen grid-less model is that in future, any high-resolution model can be extracted from the regional model to focus on any potential prospects which can further be used in Local Grid refinement modeling in Migration model by connecting this high-resolution model with regional source depocenter and remigrate the hydrocarbon at high-resolution model for de-risking.

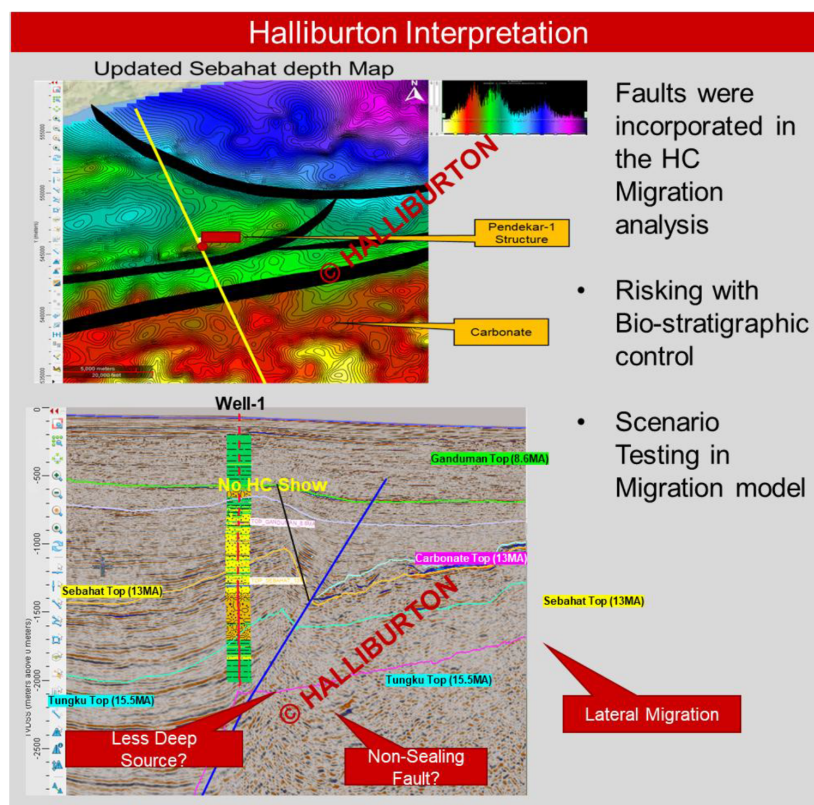


Figure 6—Prospect De-risking example of Integrating New Interpretation of Faults and Evaluating Fault Seal Scenarios with Migration Model to do the Dry Hole Analysis

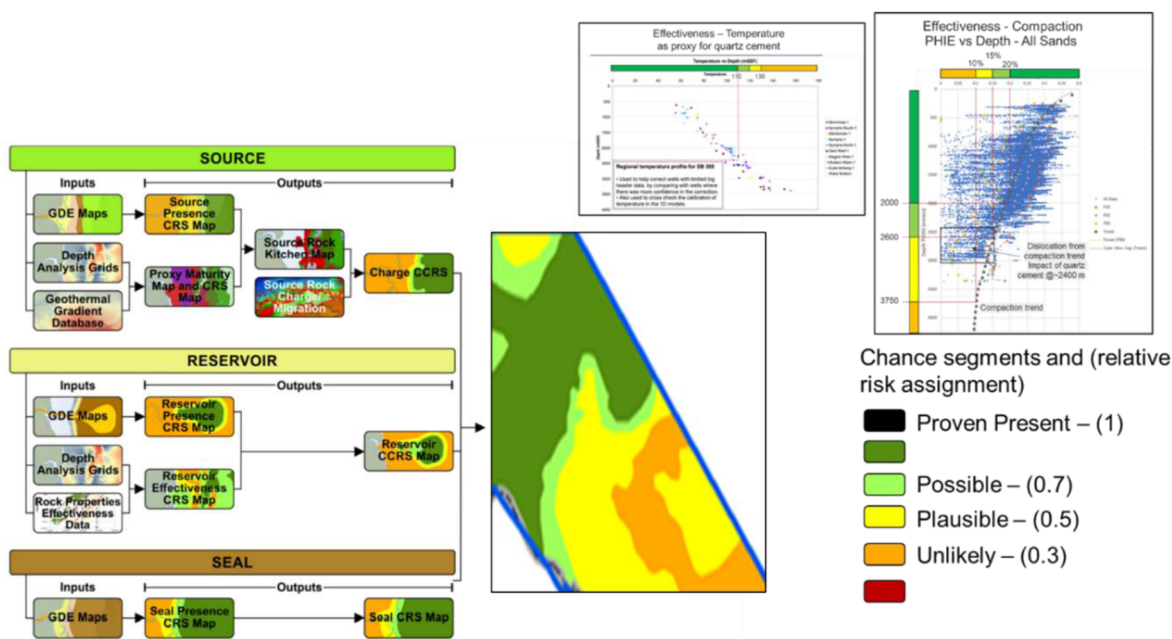
Conclusion

The integration of regional geological understanding with seismic and interpretation, which was further used as a controlling factor for region based VPC creation, is an innovative way to build 3D Geological model, even at the exploration stage with limited data control. This regional VPC-based model not only provides lateral control but also vertical connectivity as a background 3D geological trend model.

An innovative grid-less basin scale modeling approach has been adopted to address the challenges pertaining to the extent and the resolution of the model. This modeling approach considers point-based geostatistical modeling with the capability of zooming into the prospect scale without any requirement of building new model. This further helps to evaluate potential prospects at a high resolution to build the prospect inventory.

The play fairway mapping approach adopted for this study follows an industry standard Common Risk Segment (CRS) mapping approach. Within this workflow CRS maps are generated for the play-scale critical components of the petroleum system: charge, reservoir and seal. Each mapped segment is assigned a common chance value (e.g. Likely, Probable etc) and a corresponding relative risk assignment (e.g. Likely = 0.75, Probable = 0.6 etc). The schema used within this study is consistent and aligned to the schema used internally within Petronas.

The final Combined Common Chance Map (CCRS) is generated by combining the constituent CCRS maps for charge, reservoir and seal. The play fairway maps are generated by integrating results from the structural mapping (depth and isopachs) GDE mapping, basin model and the review of the existing well data.



The various trap configurations in the Sandakan Basin are illustrated schematically, based on actual hydrocarbon occurrences and potential traps that have not yet been de-risked. The primary drilling targets are the Early Miocene Tungku Clastic, Mid Miocene Sebahat Clastic and Carbonate, and Late Miocene Ganduman Clastic and Carbonate Play. While major trap styles related to structural culminations have been partially tested by drilling, several other subtle trap types, such as those associated with stratigraphic pinchouts/fans against structural highs and unconformities, remain to be explored.

The subsurface risk assignment for leads and prospects follows the Chance of Success (CoS) risk that is commonplace in the industry. In this methodology, a chance of success (CoS) (between 0 and 1) is assigned

to risk five critical geological attributes: source, timing/migration, reservoir, closure, and containment. These geological attributes have sub-components that should be considered to arrive at the chance factor. As these subcomponents are considered partially dependent the standard practice is to take the "weakest link" approach, where the lowest CoS score assigned within the subcomponents is taken as the CoS for the parent chance factor. The final CoS is derived through the multiplication of the chance factors for the five parent geological attributes. This workflow is designed so that it can be applied consistently across a range of geological settings, prospect styles and exploration maturities. The CoS can then be used as a benchmark with which to compare opportunities at the portfolio level.

The chance adequacy matrix is used to guide the assignment of CoS. This matrix considers the conclusions made from the data on the x-axis and the confidence level assigned based on the quality, quantity and control of the data and models on the y-axis. The matrix has a "bull horn" shape as the assignment of very low risk or very high risk can only be achieved where there is high confidence in the data, interpretations, and models. Likewise, with a limited or poor-quality dataset the confidence will be low and hence the assignment of very high or very low risk is not advised.

Probabilistic volume ranges are attributed to the exploration leads following an industry-standard workflow. The probabilistic volume methodology captures the inherent subsurface uncertainty within the boundary condition parameters that influence the in-place resource volumes (STOOIP or GIP).

Probabilistic ranges are defined based on user input, typically a P90 and P10 value, and then defining the distribution type (e.g., lognormal, normal, or beta). The principal inputs are area, net pay, porosity, hydrocarbon saturation, gas-to-oil ratio, formation volume factor, and recovery factor. The volume calculation methodology can be changed based on the prospect type and uncertainties.

The Probabilistic Volumetric Assessment is conducted considering the following distribution of the parameters:

- GRV: Log-Normal
- NTG: Log-Normal
- Porosity: Normal
- Oil and Gas Saturation: Normal
- Formation volume factors for OIL and GAS: Uniform
- Oil and Gas Recovery Factor: Normal

The dynamic scaling capability of the Grid-less Model has been leveraged in this Play to prospect workflow through the zooming in functionality on the individual prospects and running grid-less volumetrics without any requirement of building separate siloed model for multiple prospects. The model has the capability to add new points in the prospect area which further get recomputed in terms of reservoir facies and other volumetrics properties with the variogram and the data influence from a larger area in order to maintain the consistency in the model. This workflow makes this model more of an evergreen subsurface google earth which can be updated over future with any new data incorporation with the flexibility of resolution and extent.

References

1. Haile, N.S. and Wong, N.P.Y., 1965. The geology and mineral resources of the Sandakan area and parts of the Kinabatangan and Labuk Valleys, Sabah. *Geological Survey Borneo Region*. Malaysia, Memoir 16

2. Haile, N.S., 1969. Geosynclinal theory and the organizational pattern of the Northwest Borneo Geosyncline, *Quarterly Journal Geological Society of London*, **124**, 171–194
3. Haile, N.S., 1973. The recognition of former subduction zones in Southeast Asia, In: Tarling, D.H. and Runcorn, S.K., eds. "*Implications of Continental Drift to the Earth Sciences*". 2. Academic Press, London, 885–892
4. Hamilton, W., 1979. Tectonics of the Indonesian region. US Geological Survey Professional Paper 1078
5. Hazebroek, H.P. and Tan, D.N.K., 1993. Tertiary tectonic evolution of the NW Sabah continental margin. In: Teh, G.H., ed., Proc. Symp Tectonic Framework and Energy Resources of the Western Margin of Pacific Basin. Bulletin of the Geological Society of Malaysia, **33**, 195–210
6. Hinz, K. and Schluter, H.V., 1985. Geology of the Dangerous Ground, South China Sea and Continental Margin off SW Palawan: *Results of SONNE Cruise SO-23 and SO-27, Energy* **10**, 297–315
7. Hinz, K., Kempter, E.H.K. and Schluter, H.U. 1985. The Southern Palawan-Balabac area: an accreted or non-accreted terrane? ASCOPE Proceedings. **2**. 48–72
8. Hinz, K., Fritsch, J., Kempter, E.H.K., Mohammad, A.M., Meyer, J., Mohamed, D., Vosberg, H., Weber, J. and Benavides, J. 1989. Thrust tectonics along the North-western Continental Margin of Sabah/Borneo. *Geologische Rundschau Band* 78. Helt 3
9. Hinz, K., Block, M., Kudrass, H.R. and Meyer, H., 1991. Structural elements of the Sulu Sea, Philippines Geol. *Jahrb. Reihe, A*, **127**, 183–506
10. Hutchison, C.S., 1975. Ophiolites in Southeast Asia. *Bulletin of the Geological Society of America*, **86**, 797–806
11. Hutchison, C.S., 1988. Stratigraphic tectonic model for eastern Borneo. *Bulletin of the Geological Society of Malaysia*, **22**, 135–151
12. Hutchison, C.S., 1992a. The Southeast Sulu Sea, a Neogene marginal basin with outcropping extensions in Sabah. *Bulletin of the Geological Society Malaysia*, **32**, 89–108
13. Hutchison, C.S. 1992b. The Eocene unconformity in Southeast Asia and East Sundaland. *Bulletin of the Geological Society of Malaysia*, **32**, 69–88
14. Hutchison, C.S., 1996. The 'Rajang accretionary prism' and 'Lupar Line' problem of Borneo. In: Hall R. and Blundell, D. (eds.). *Tectonic Evolution of Southeast Asia*, Geological Society of London Special Publication, **106**, 247–261
15. Hutchison, C.S., 1997. Tectonic framework of the Neogene basins of Sabah. Abstracts Geological Society of Malaysia Petroleum Geology Conference 1997
16. Idris, M.B., and Kok, K.H., 1990. Stratigraphy of the Mantanani Islands, Sabah. *Bulletin of the Geological Society of Malaysia*, **26**, 35–46.
17. Irving, E.M., 1949. Notes on the geology of Balabac Island, Palawan, *Philippines Island Philippines Geologist*, **3**. 14–21.
18. Ismail Che Mat Zin, 1994. Dent Group and its equivalent in the offshore Kinabatangan area, East Sabah. *Bulletin of the Geological Society of Malaysia*. **36**, 127–143.
19. MPM 2023. East Sabah Tectonostratigraphic and Carbonate Play Evaluation.